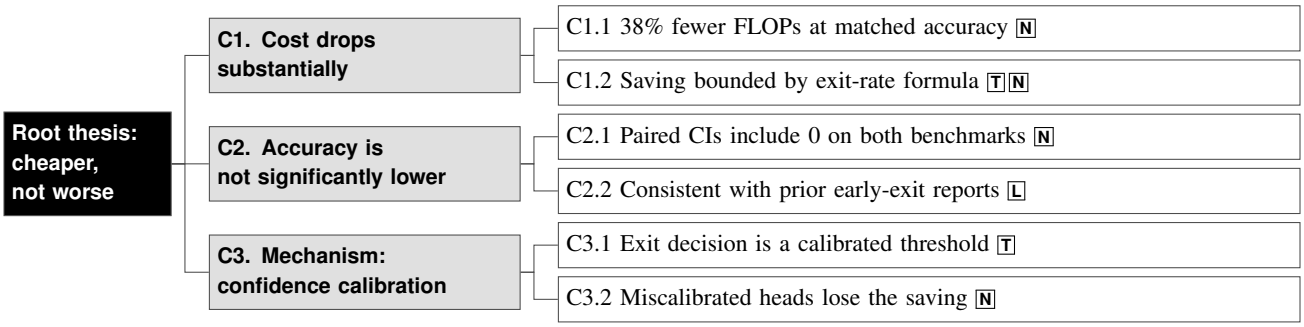
Argument Tree: «Short Title of the Work»

«Target venue or purpose» · «scope, e.g. datasets» · status «date»

Evidence types: T theory (formally checked where marked) L literature N numerical (this study)

ROOT THESIS

«One to three sentences. The single claim the whole document defends, scoped exactly to what the evidence supports. Bold the two or three load-bearing phrases.» Example: A **lightweight early-exit head** cuts inference cost by **38%** on two image benchmarks with **no significant accuracy loss**; the saving is explained by a simple confidence bound.



C1. Inference cost drops substantially

«One sentence restating the claim precisely.» The early-exit head removes 38% of FLOPs at matched accuracy.

THEORETICAL EVIDENCE

P1 («formal name, e.g. Lean theorem or lemma number»): expected cost $= c_{\text{head}} + (1 - r)\, c_{\text{full}}$, where r is the exit rate. «State what was proved, under which assumptions, and how it was checked.»

NUMERICAL EVIDENCE

C1.1 «Benchmark A»: 38% fewer FLOPs (5 seeds, SD 1.2); «Benchmark B»: 35% (5 seeds, SD 2.0).
C1.2 Observed exit rate $r = 0.41$; predicted saving 37%, measured 38%.

CAVEATS AND BOUNDARIES

Savings measured on one GPU type at batch size 1; larger batches reduce the benefit.

C2. Accuracy is not significantly lower

Best configuration vs best configuration, seeds aggregated within each.

NUMERICAL EVIDENCE

Benchmark	seeds	baseline	ours	Δ	95% CI
«A»	5	91.2±0.4	90.9±0.5	−0.3	[−0.9, +0.3]
«B»	5	78.5±0.8	78.6±0.7	+0.1	[−0.8, +1.0]

LITERATURE EVIDENCE

C2.2: Early-exit studies [1] report small accuracy changes at similar exit rates. «Say exactly what each cited work shows and how it relates; name the closest prior work and the precise difference.»

CAVEATS AND BOUNDARIES

"Not significantly lower" is not equivalence; no equivalence test was run.

C3. Mechanism: confidence calibration

The saving depends on the exit head being calibrated.

THEORETICAL EVIDENCE

P2: if the head is calibrated, thresholding confidence at τ bounds the exit error by $1 - \tau$. «Proof status.»

NUMERICAL EVIDENCE

C3.2: with temperature scaling removed, the exit rate falls to 0.12 and the saving to 9%.

GLOBAL LIMITATIONS AND SOURCES

«Statistical limits (which differences are significant, seed counts), scope limits (datasets, models), which claims are proved vs numerical only, citation-verification date, and where the raw results live.»

References (verified «date»)

[1] «Authors. Title. *Venue* Year. arXiv/DOI.»